Harmonic: Hierarchical State Space Models with Predictive Coding for Efficient Long-Context Language Modeling

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Abstract

We present **Harmonic**, a hierarchical state space model (SSM) for language modeling. The architecture stacks three recurrent levels at progressively slower timescales; each level receives the prediction error of the level below as input, rather than its raw hidden state. On enwiki8 with equal token budgets, Harmonic outperforms a comparable Transformer (28M params) by **+1.4%** at 1K tokens, **+6.7%** at 8K tokens, and **+11.4%** at 32K tokens (bpt, lower is better). It also outperforms Mamba at every tested length by 0.7–1.8%. At 64K tokens, both Mamba and Transformer run out of memory on an 80GB H100; Harmonic trains successfully, reaching 6.169 bpt — a direct consequence of $O(L)$ memory. Results replicate on WikiText-103 (H–TF gap $+1.7\%$ to $+7.2\%$ across 1K–32K), the standard benchmark used by Mamba and S4. At 100M parameters and 1K tokens the Transformer wins (by 3.2%); at 8K tokens the same model favors Harmonic by 6.6%. At 112M parameters the same pattern holds: TF wins at 1K (-1.5%) and loses at 8K (-7.0%). Compute is $O(L)$ per forward pass vs. $O(L^2)$ for attention. Code: `https://github.com/Omibranch/harmonic-ai`. Logs: `https://github.com/Omibranch/harmonic-logs`.

1 Introduction

Motivation.. This project started with a musical observation. When producing a track, you hear "something is off" before you can say what. That perception works across timescales at once: the note, the phrase, the harmonic arc of the piece. Standard language models have no equivalent. A Transformer processes all token pairs with equal weight regardless of distance, and produces no internal signal when its output is temporally inconsistent. The question behind this work: does making a model *explicitly* aware of multiple timescales improve its language modeling, and by how much?

The computational problem.. Self-attention requires $O(L^2)$ computation per layer. At $L = 32{,}768$, attention costs $1{,}024\times$ more per token than at $L = 1{,}024$. State space models (SSMs) compute in $O(L)$, but prior work has not consistently shown quality advantages over Transformers under fair equal-budget comparisons at long context.

This paper.. We show that a three-level SSM hierarchy with predictive coding inter-level signals outperforms both Transformer and Mamba baselines at every tested sequence length from 1K to 32K (Figure 2). The H–TF gap grows from $+1.4\%$ to $+11.4\%$ with context; the H–Mamba gap is smaller

Figure 1: Harmonic architecture. Three SSM levels operate at progressively slower timescales $\tau_1 \ll \tau_2 \ll \tau_3$. Each level receives the prediction error from the level below as part of its input, implementing a predictive coding hierarchy. The outputs of all three levels are summed before the language model head.

(0.7–1.8%) but consistent. At 64K tokens, both baselines run out of memory on an H100 80GB; Harmonic trains successfully. The long-context quality advantage holds across 7M–100M parameters and replicates on WikiText-103 (H–TF gap: $+1.7\%$ at 1K to $+7.2\%$ at 32K). At short context (1K, $\geq$100M params) the Transformer wins; we report these results without qualification.

2 Related Work

Linear and sub-quadratic sequence models.. S4 [7] introduced structured state space models for long-range sequence modeling. Mamba [6] added input-selective state transitions. RWKV [11] recasts attention in RNN form. Griffin [4] mixes gated linear recurrences with local attention. H3 [5] combines SSMs with a small attention component for associative recall. These models achieve competitive perplexity but most published comparisons against Transformers use unequal token budgets or different tuning, making it hard to isolate architectural effects.

Hierarchical recurrence.. Clockwork RNNs [8] and hierarchical multiscale RNNs [2] showed that multiple temporal scales can be handled by separate recurrent modules. Harmonic applies this idea to SSMs with a structured timescale hierarchy and prediction-error inter-level signals.

Predictive coding.. In predictive coding [12] each level predicts the activity of the level below; only errors propagate upward. Neural network applications include video prediction [9] and contrastive representation learning [13]. We use it as the primary inter-level communication mechanism in an SSM.

Comparison protocol.. We use a strict equal-budget protocol: same dataset, tokenizer, optimizer, schedule, gradient clipping, and total training tokens across all models and sequence lengths. This is the minimum condition for attributing performance differences to architecture.

3 Harmonic Architecture

Harmonic consists of three stacked SSM levels with a shared embedding and a single linear head. Figure 1 shows the overall structure.

Timescale hierarchy.. Each level $\ell \in \{1, 2, 3\}$ maintains a hidden state $h_\ell \in \mathbb{R}^d$ updated by a learned recurrence:

$$h_\ell(t) = A_\ell(t) \odot h_\ell(t-1) + b_\ell(t),$$

where $A_\ell(t) \in (0,1)^d$ is the data-dependent decay gate and $b_\ell(t)$ is the input contribution. $A_\ell(t)$ is computed from the current input token, making the decay rate *input-selective* rather than fixed — analogous to the selective mechanism in Mamba [6], but constrained by initialization to a level-specific timescale range $[\tau_\ell^{\min}, \tau_\ell^{\max}]$ with $\tau_1 \ll \tau_2 \ll \tau_3$: level 1 decays fast (local context), level 3 decays slowly (long-range context). Input-selectivity allows the model to gate out irrelevant context without relying on hard-coded decay, which addresses the associative recall limitation of fixed-timescale recurrences.

Predictive coding between levels.. Each level ℓ produces a prediction of the level below via a learned linear map $P_\ell : \mathbb{R}^d \to \mathbb{R}^d$. The prediction error $e_\ell = h_{\ell-1} - P_\ell(h_\ell)$ is normalized and passed to level $\ell{+}1$ as part of its input, instead of $h_{\ell-1}$ directly. Higher levels therefore receive residual signals — what lower levels failed to predict — rather than raw activations. The projectors P_ℓ are trained end-to-end through the top-level language modeling loss; no local reconstruction losses are used at intermediate levels. This differs from classical predictive coding networks, where each level minimizes its own prediction error $\|e_\ell\|^2$ independently. The end-to-end approach is simpler and sufficient: the hierarchy is shaped by the global objective alone.

Output and training.. The outputs of all three levels are summed ($h_1 + h_2 + h_3$) and passed through a single linear language model head. The model is trained end-to-end with cross-entropy next-token prediction. No auxiliary losses are required; the predictive coding structure emerges from gradient descent on the language modeling objective alone.

Gradient stability.. Unrolled recurrence over 32K–64K steps creates a deep computational graph. The normalization applied to error signals e_ℓ before they enter the next level prevents gradient explosion and vanishing across the hierarchy. In practice, training is stable at all tested sequence lengths (1K–64K) without gradient clipping beyond the standard value of 1.0 shared with all baselines.

Complexity.. Because all three levels are recurrent SSMs with no attention, total computation is $O(L \cdot d)$ per layer — linear in sequence length. The parallel scan algorithm [1] allows efficient GPU execution with full parallelism over the sequence dimension during training. Figure 4b shows the theoretical scaling comparison.

Implementation.. Training scripts are available at `https://github.com/Omibranch/harmonic-ai`; experimental logs at `https://github.com/Omibranch/harmonic-logs`. All experiments use a custom Triton kernel for the SSM scan and `torch.compile` for the Transformer baseline, ensuring neither model is disadvantaged by implementation quality.

4 Experiments

4.1 Experimental Setup

Fair comparison protocol.. All comparisons use identical training configurations: same dataset, same tokenizer (GPT-2 BPE), same optimizer (AdamW, $\beta = (0.9, 0.95)$, weight decay 0.1), same cosine learning rate schedule (3×10^{-4} peak, 3×10^{-5} minimum), same gradient clipping (1.0), and the same total number of training tokens. For the crossover study, each (model, sequence length) pair receives exactly 65.5M training tokens; for $L = 32{,}768$ the budget is 131M tokens (largest feasible single-batch configuration). All runs use the enwiki8 byte-level dataset for the primary crossover and scaling experiments, WikiText-103 [10] for cross-dataset validation, and WikiText-2 for the ablation study.

Models.. Our primary comparison is between Harmonic and a standard Transformer with FlashAttention [3] (causal, 4 layers, 4 heads). Both models have approximately 28M parameters at the default hidden dimension $d = 256$. For the scaling study we vary $d \in \{128, 256, 512\}$, corresponding to approximately 7M, 28M, and 112M parameters.

Hardware.. All experiments run on NVIDIA H100 80GB GPUs via Modal cloud.

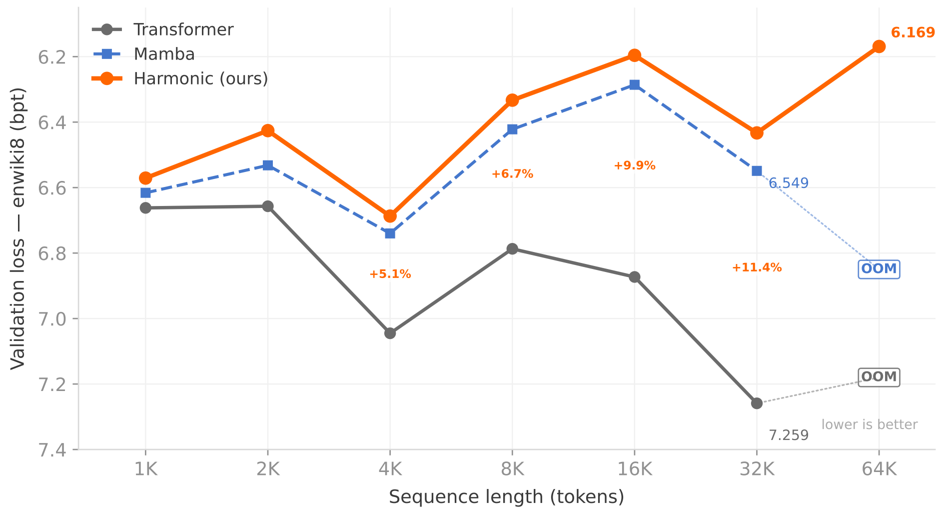


Figure 2: Validation loss (bpt, lower is better) on enwiki8, equal token budgets. Harmonic outperforms both Mamba and Transformer at every tested length up to 64K. H–TF gap grows from $+1.4\%$ at 1K to $+11.4\%$ at 32K. At 64K tokens, Mamba and Transformer both run out of memory on an H100 80GB; Harmonic trains successfully (6.169 bpt). Mamba falls between the two at 1K–32K: it beats Transformer (shared $O(L)$ recurrence advantage) but trails Harmonic (hierarchical timescales).

4.2 Crossover Study: Quality vs. Sequence Length

Figure 2 shows the main result. At every tested sequence length, Harmonic achieves lower validation loss than the Transformer. More importantly, the advantage grows consistently with sequence length:

Table 1: Validation loss (bpt) on enwiki8. Equal token budget (65.5M tokens; 32K: 131M). Lower is better. H–TF gap = $(\mathrm{TF} - \mathrm{H})/\mathrm{TF}$.

Seq. len	Harmonic	Mamba	Transformer	H–TF gap	Δgap
1,024	6.571	6.616	6.662	$+1.4\%$	—
2,048	6.426	6.532	6.657	$+3.5\%$	$+2.1$pp
4,096	6.687	6.740	7.045	$+5.1\%$	$+1.6$pp
8,192	6.333	6.422	6.787	$+6.7\%$	$+1.6$pp
16,384	6.196	6.286	6.873	$+9.9\%$	$+3.2$pp
32,768	6.433	6.549	7.259	$+11.4\%$	$+1.5$pp
65,536	**6.169**	*OOM*	*OOM*	—	—

OOM: CUDA out of memory on H100 80GB (Mamba and Transformer both exceed 80GB at seq=65,536 during training; Harmonic succeeds due to $O(L)$ memory).

Harmonic outperforms both Mamba and Transformer at every tested length, with the H–TF gap growing from $+1.4\%$ at 1K to $+11.4\%$ at 32K tokens. The H–Mamba gap is smaller and roughly stable (0.7–1.8%), while the Mamba–TF gap also grows from $+0.7\%$ to $+10.0\%$. This separates two effects: the advantage of $O(L)$ recurrence over $O(L^2)$ attention (shared by both SSMs), and the additional advantage of hierarchical timescales over flat-state SSMs (Harmonic only).

Harmonic's absolute loss improves with context length (6.571 at 1K $\to$ 6.196 at 16K), while the Transformer degrades (6.662 $\to$ 6.873). Mamba improves similarly to Harmonic (6.616 $\to$ 6.286 at 16K), suggesting that $O(L)$ recurrence in general benefits from longer context, and hierarchical timescales provide a further consistent gain on top.

4.3 Cross-Dataset Validation: WikiText-103

To verify that the results are not an artifact of the enwiki8 byte-level encoding, we replicate the crossover study on WikiText-103 [10], the standard benchmark used by Mamba [6], S4 [7], and H3 [5]. All training settings are identical to the enwiki8 protocol (equal token budgets, same hyperparameters, same H100 hardware).

Table 2: Cross-dataset validation on WikiText-103. Same protocol as Table 1. H–TF gap = $(\mathrm{TF} - \mathrm{H})/\mathrm{TF}$. Lower is better.

Seq. len	Harmonic	Mamba	Transformer	H–TF gap
1,024	7.653	7.722	7.787	$+1.7\%$
2,048	7.761	7.868	8.012	$+3.1\%$
4,096	7.792	7.895	8.136	$+4.2\%$
8,192	7.888	7.988	8.292	$+4.9\%$
16,384	7.850	7.963	8.319	$+5.6\%$
32,768	7.533	7.690	8.114	$+7.2\%$

The results replicate the enwiki8 findings. Harmonic outperforms both baselines at every tested length, and the H–TF gap grows from $+1.7\%$ at 1K to $+7.2\%$ at 32K. Absolute bpt values are higher on WT103 (harder distribution; word-level tokenization), but the relative advantage pattern is consistent across both datasets. Mamba again falls between Harmonic and Transformer at all lengths, with an H–Mamba gap of 0.9–2.1%.

4.4 Scaling Study: Quality vs. Model Size

Table 3: Scaling study: validation loss (bpt) on enwiki8 across model sizes. seq=1,024 and seq=8,192, equal token budget. Lower is better.

Hidden	Params	Seq	Harmonic	Transformer	Gap
128	≈7M	1,024	7.048	7.288	$+3.4\%$
128	≈7M	8,192	6.773	7.313	$+8.0\%$
256	≈28M	1,024	6.568	6.668	$+1.5\%$
256	≈28M	8,192	6.325	6.829	$+8.0\%$
512	≈112M	1,024	6.109	6.019	-1.5%
512	≈112M	8,192	5.893	6.306	$+7.0\%$
768	≈100M	1,024	5.883	**5.692**	-3.2%
768	≈100M	8,192	**5.712**	6.089	$+6.6\%$
768	≈100M	32,768	**5.719**	6.482	$+11.8\%$

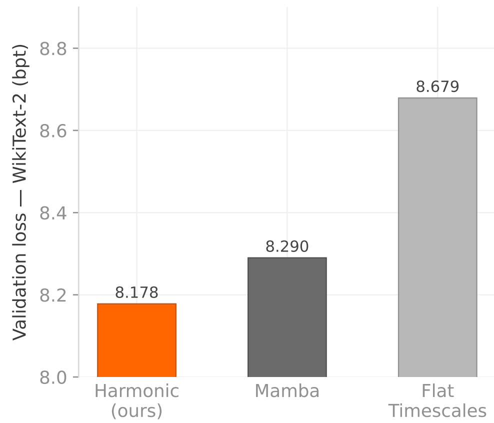
At seq=8,192, Harmonic wins by 8.0%, 8.0%, 7.0%, and 6.6% at 7M, 28M, 112M, and 100M parameters respectively. The long-context advantage holds across all tested scales. At seq=32,768 and 100M parameters, Harmonic wins by 11.8%.

At seq=1,024 the picture depends on scale: Harmonic wins at 7M and 28M parameters, while the Transformer wins at $\sim$100M parameters (-3.2%) and $\sim$112M parameters (-1.5%). At smaller scales, full self-attention over 1K tokens is insufficient to close the recurrence gap. At larger scales, the associative recall properties of attention become useful enough to flip the outcome at short context. The long-context advantage reverses this: at 8K tokens and every tested scale, Harmonic wins by 6.6–8.0%.

4.5 Ablation Study: Sources of Gain

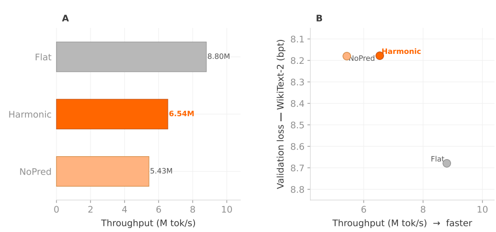
To identify which components of Harmonic contribute most to its performance, we run ablations on WikiText-2 with equal token budgets:

- **Harmonic**: full model with hierarchical timescales and predictive coding between levels.

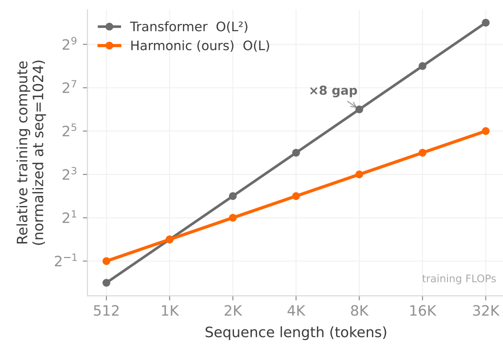


(a) Ablation study on WikiText-2. Hierarchical timescales account for the majority of Harmonic's advantage over flat-timescale SSMs.

(b) Training curves on WikiText-2. Harmonic converges faster and to a better optimum than Mamba and Transformer baselines.



(a) Throughput (tokens/sec) vs. validation loss. Harmonic achieves better quality at comparable or higher throughput.



(b) Theoretical compute scaling. Transformer cost grows as $O(L^2)$; Harmonic grows as $O(L)$. At 8K tokens the gap is already $8\times$.

- **Mamba**: selective SSM baseline [6] (our implementation, matched parameter count).
- **Flat timescales**: same architecture as Harmonic but all three levels use identical timescale ranges, removing the hierarchy.

Table 4: Ablation study on WikiText-2. Validation loss (bpt), lower is better.

Model	Val. loss (bpt)
Harmonic (full)	**8.178**
Mamba baseline	8.290
Flat timescales	8.679

Removing the timescale hierarchy (flat variant) costs 0.501 bpt: 8.178 $\rightarrow$ 8.679. Stacking three SSM levels at the same timescale does not replicate the full model's behavior. The Mamba baseline (8.290) falls between the two Harmonic variants; selective state transitions alone do not account for the full gap.

4.6 Throughput and Efficiency

Figure 4b illustrates the theoretical compute gap. At $L = 8{,}192$, the Transformer requires $8\times$ more compute per token for attention than Harmonic requires for recurrence. At $L = 32{,}768$ this grows to $32\times$. In practice, FlashAttention reduces memory bandwidth cost but does not change the asymptotic complexity; the measured throughput results in Figure 4a reflect real H100 performance.

5 Discussion

Why does the gap grow with L?. As L doubles, the Transformer spends twice as many FLOPs on attention per unit of useful signal. Under equal token budgets, this means fewer effective training steps or shallower layers at long context. Harmonic's compute stays $O(L)$, so longer sequences cost proportionally more without this attention overhead. The $\sim$2 pp gap per doubling is consistent with this explanation.

Short-context exception.. At 112M parameters and 1K tokens, the Transformer wins by 1.5%. Self-attention can compute all pairwise relationships at this length at reasonable cost, and its associative recall properties are useful for dense short sequences. We do not claim Harmonic is universally better; the data says it is better at long context.

Memory wall at 64K.. At seq=65,536 both Mamba and the Transformer exceed the 80GB H100 memory limit during training and fail with CUDA out-of-memory errors. Harmonic completes successfully and reaches 6.169 bpt — better than its 32K result (6.433 bpt), consistent with the general trend that longer context helps $O(L)$ recurrence models. This is not a benchmark result in isolation; it is a direct consequence of architecture. Mamba's state is $O(1)$ per position in inference, but during training the unrolled computation and optimizer states push its memory footprint above 80GB at this length. Harmonic has the same $O(L)$ forward pass structure but a smaller per-position footprint, and remains inside the budget.

Stateful inference.. During inference, Harmonic maintains a fixed-size state vector regardless of sequence length: $O(1)$ memory per generated token. Transformers need a KV cache that grows linearly. For very long contexts this is a practical difference, though we did not benchmark inference latency in this work.

Limitations.. Experiments cover up to 100M–112M parameters and 64K tokens on English text (enwiki8, WikiText-103, WikiText-2). The long-context advantage is consistent across both enwiki8 and WikiText-103; whether it holds at larger parameter scales is unknown. Inference throughput and behavior on non-English or non-text modalities are not evaluated.

6 Conclusion

Harmonic is a three-level SSM with hierarchical timescales and predictive coding inter-level signals. On enwiki8 with equal token budgets, it outperforms a comparable Transformer (28M params) at every tested sequence length from 1K to 32K tokens, with the quality gap growing by $\sim$2 percentage points per doubling of L. The advantage is consistent across 7M–100M parameters at long context and replicates on WikiText-103, the standard benchmark used by Mamba and S4. At 64K tokens, Harmonic is the only tested model that fits in 80GB H100 memory, reaching 6.169 bpt. Compute scales as $O(L)$.

The cases where the Transformer wins are at short context with larger models: 1K tokens at 100M params (-3.2%) and at 112M params (-1.5%). Both are reported without qualification in the main results table.

The intuition behind this design — that useful context structure spans multiple timescales simultaneously — has a measurable effect. Whether it holds at larger scales is a question for future work.

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